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ASSIGNMENT - 1

Q An LSTM cell has current ..... Calculate the updated cell state & final hidden state for this time stamp.

Given,

$c_{t-1} = 4$ $\quad h_{t-1} = 0.5$

$f_t = 0.7$ $\quad o_t = 0.6$

$i_t = 0.5$ $\quad \tilde{c}_t = 2.4$

Updated Cell State:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$
$$= (0.7 \times 4) + (0.5 \times 2.4)$$
$$= 2.8 + 1.2$$
$$\boxed{c_t = 4.0}$$

Final hidden state:

$$h_t = o_t \odot \tanh(c_t)$$
$$= 0.6 \times \tanh(4.0)$$
$$= 0.6 \times 0.9993 \quad [\tanh(4) \approx 0.9932]$$
$$= 0.59958$$
$$\boxed{h_t \approx 0.6}$$

Bharti

Teacher's Signature _____